Probabilistic modeling & experimental pragmatics

Towards a computational cognitive science of communication

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Abstract

Probabilistic modelling has strongly contributed to the success of experimental pragmatics by providing a formal-computational bridge between theoretical concepts, on the one hand, and, on the other hand, direct, even quantitative predictions, for the likelihood of single observations in empirical data sets. This overview summarizes the main driving idea behind probabilistic pragmatics, namely to explain the mutually attuned speaker production and listener interpretation policies as a form of resource-efficient, approximately optimal adaptation to the requirements of linguistic communication. The role of probabilistic models as theory-driven statistical models is sketched, and a structured overview of the recent literature is provided. Looking forward, an argument is developed in support of a cognitive turn in experimental pragmatics towards a computational cognitive science of communication; an enterprise which, as argued here, entails developing models of polices that are adapted to the complex "mind-dynamics" required for dealing with uncertainty about the conversational context and common ground in an open-ended setting.

1 Experimental pragmatics and computational modeling

Pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the philosophy of language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018).[1]

The experimental turn in pragmatics was accompanied by formal and computational work from the start, with two main strands of approaches emerging at opposite ends of a spectrum. On one end of the spectrum, *grammaticalism* focuses on language as an abstract, computationally encapsulated system (e.g., Chierchia, Fox, and Spector, 2012). Grammaticalism strives to give formally precise explanations for how pragmatic inferences (focusing almost exclusively on interpretation) could arise in a language-specific computational module. On the other end of the spectrum, *probabilistic pragmatics* seeks explanations of pragmatic phenomena from the point of view that linguistic

[1]This development paralleled similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

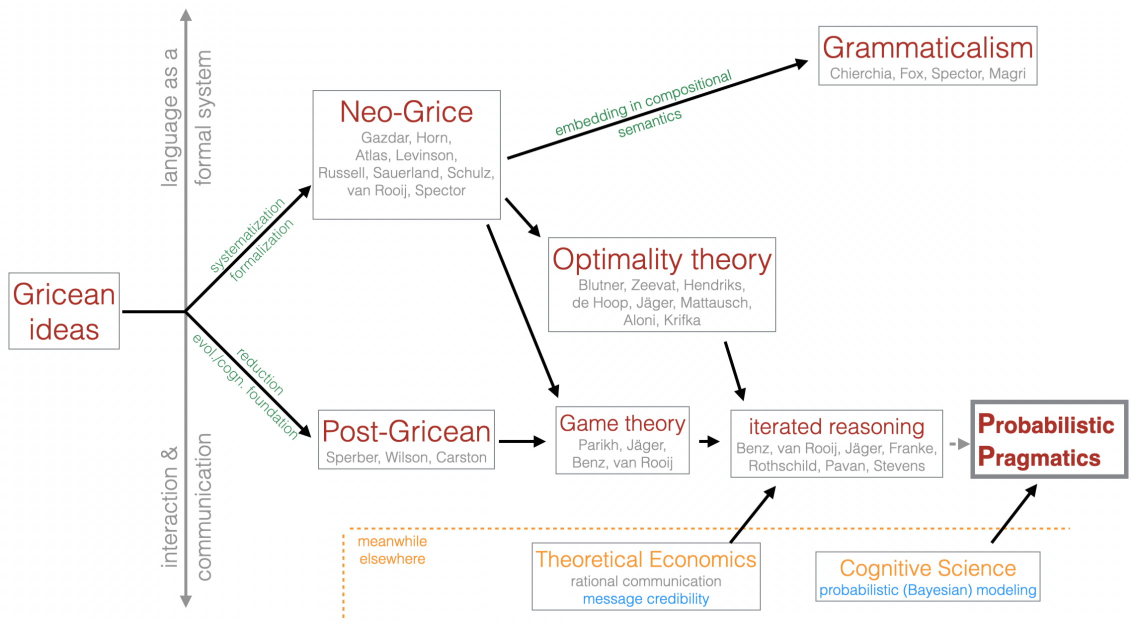

Figure 1: Genealogy of probabilistic pragmatics. Starting from the foundational work of Paul Grice, the visualization shows the rough historical development (from left to right) of different approaches that elaborated on Grice's work. These approaches can be roughly situated along an axis of abstraction, looking at "language as a formal system" vs. looking at "linguistic communication." Probabilistic pragmatics is situated more towards the latter, and can be seen as the product of converging ideas traceable through Post-Gricean approaches, game- and optimality-theory, but also including general ideas on resource-efficient approximately optimal polices from cognitive science. For different approaches, names of key contributors are named without any claims regarding exhaustiveness or implications of their particular contributions.

communication is a special form of social interaction, also taking aspects of domain-general cognition of language users into account (e.g., Frank and Goodman, 2012; Goodman and Stuhlmüller, 2013; Degen, 2023). Figure 1 gives a coarse-grained overview of the development of, in particular, probabilistic pragmatics as the culmination point of research that combines aspects of individual cognition, such as (resource-)bounded rationality (Griffiths, Lieder, and Goodman, 2015; Lieder and Griffiths, 2019, e.g.,), and functional explanations based on optimal information transfer in communication (e.g., Franke, 2017; Stevens and Benz, 2018).

While experimental data can mitigate between alternative explanations proposed in the grammaticalist tradition, I argue here that the more socio-cognitive approach adopted by probabilistic pragmatics is, if done right, much better suited to incorporate *both* linguistic theory *and* aspects of general cognition (such as attention, meta-cognition, or memory), as well as the concrete affordances of a conversational situation or an experimental task, into a unified computational model of the *data-generating process*, i.e., the (cognitive) processes by which individual data observations, such as concrete data points recorded from an experiment, come about. Seen in this light, computational modeling in pragmatics can serve a bridging function between language, linguistic cognition and domain-general aspects of information-processing and decision-making. The central argument of this paper is, consequently, that the momentum gained from the experimental turn should be channeled into a dedicated *cognitive turn*, which seeks to explain language use as the intricate interplay between language-specific and domain-general cognition. Computational cognitive-pragmatic models play a pivotal role in this cognitive turn, as they require detailed commitment about the role of linguistic theory (latent, established concepts) in the data-generating process, and the way that general cognitive factors shape linguistic behavior. As I will argue here, recent work in modelling-oriented experimental pragmatics has started to "go cognitive" in this sense, but that a more dedicated embrace of the cognitive turn is needed. Looking forward, I sketch possible future directions.

The paper is structured as follows. Section 2 introduces probabilistic models pragmatic choice and reasoning in the (widely conceived) Rational Speech Act tradition and the notion of resource-efficient approximately optimal policies. Section 3 elaborates on the idea of cognitive models acting as theory-driven statistical models for the data-generating process. Section 4 develops an argument that, in order for probabilistic models or pragmatic reasoning to become more ecologically plausible, we need to understand how resource-efficient, approximately optimal polices are adapted to the complex "mind-dynamics" entailed by probabilistic reasoning about context and common ground in an open-ended setting, which entails accommodating general cognitive processes, such as attention, awareness and meta-cognition.

2 Probabilistic pragmatics

Notable early probabilistic approaches to pragmatics include pioneering work by Merin (1994, 1999), Zeevat (2014), and Russell (2012). A particularly prominent style of probabilistic modeling in pragmatics is the Rational Speech Act framework (Frank and Goodman, 2012; Scontras, Tessler, and Franke, 2021; Degen, 2023), the basic set-up of which (as detailed below) was anticipated in decision-theoretic work by Anton Benz (Benz, 2006; Benz and van Rooij, 2007).[2] Early contributions in the evolving RSA tradition looked at implicature (Bergen, Levy, and Goodman, 2012; Goodman and Stuhlmüller, 2013), stylized reference games (Frank and Goodman, 2012; Degen and Franke, 2012; Degen, Franke, and Jäger, 2013) , or the place of probabilistic notions in semantics and pragmatics in general (Goodman and Lassiter, 2014).

What unites probabilistic approaches to pragmatics is a focus on explaining pragmatic phenomena in terms of what agents *do* in linguistic communication. More concretely, following the general tenets of Gricean (Grice, 1975) and Post-Gricean pragmatics (Sperber and Wilson, 1995; Wilson and Sperber, 2004; Blutner and Zeevat, 2004), an emergent view in concurrent cognitively oriented pragmatic modeling is that linguistic behavior is to be explained as **REAL behavior** (**r**esource-**e**fficient, **a**pproximately optima**l**). The assumption of (approximate) optimality of behavior allows for functional-teleological explanations of *why* a particular behavior or general phenomenon is likely to be supported as a *good* solution to a recurring communication problem (e.g., Nowak and Krakauer, 1999; Skyrms, 2010). Resource-efficiency stresses that realizations of implementations of (approximately) optimal solutions may trade-in some ulterior success for easier representation or policy execution (e.g., Lieder and Griffiths, 2019; Lai and Gershman, 2024).

The assumption that linguistic communication is based on REAL behavior leads to a picture of pragmatic interpretation as a form of inverse, or abductive, reasoning (Hobbs, Stickel, and Martin, 1993; Bunt and Black, 2000; Franke and Jäger, 2016), implemented as Bayesian inference. Since

[2] Earlier (e.g. Kipper and Jameson, 1994) and independent later work (e.g., Tellex et al., 2014) also used the literal-interpretation → pragmatic speaker → Bayesian interpreter characteristic of (vanilla) RSA models.

it is natural to assume that a speaker's linguistic behavior (their choice of words, the fact that they spoke in the first place, their intonation, or speech-accompanying gesture, etc.) is at least partly determined by a latent, i.e., unobservable, mental state of the speaker, the main task of pragmatic interpretation is to reason backwards, as it were, and compare the relative likelihood (and prior plausibility) of different conceivable causes of the speaker's behavior. Bayesian probabilistic inference provide the normative framework for this kind of inverse reasoning. Given a probabilistic speaker policy $\pi(a \mid t)$, which specifies how likely a speaker is to perform an action $a \in A$ when in state $t \in T$, together with a prior distribution over how likely different states t are, the normatively correct way of updating these prior beliefs is given by Bayes' rule as follows:

$$P(t \mid a) = \frac{\pi(a \mid t)\ P(t)}{\sum_{t' \in T} \pi(a \mid t')\ P(t')} \tag{1}$$

To add content to this basic—and, at least as a starting point, arguably natural—picture of interpretation, at least three things need to be specified: (i) the relevant set of states T; (ii) the relevant set of actions A; and (iii) the speaker's behavioral policy π. This is best illustrated with a simple example of referential communication. The states T are potential referents in a perceptual context shared between speaker and listener. The (linguistic) actions A are a finite set of utterances or descriptions, identified, say, at the level of surface form, such as "the blue triangle." The speaker's policy π can then be defined on the basis of standard assumptions about how speakers are expected to behave in situations of the relevant kind, for example, as envisaged by Grice's Maxims of Conversation (Grice, 1975). In particular, we can define a probabilistic policy in terms of a utility function that scores each action (utterance) a in a given state t based on a's truth-value in t (quality), the degree to which a conveys information about t as opposed to any other $t' \in T$ (informativity), and the cost $C(a)$ of performing a:

$$\mathrm{U}(a, t) = \mathrm{Informativity}(a) + \mathrm{Truth}(a, t) + \mathrm{Cost}(a) \tag{2}$$

The speaker's policy is then defined as (approximately) optimal behavior for this utility function, for instance as implemented via the softmax function (Franke and Degen, 2023):

$$\pi(a \mid t) = \frac{\exp \alpha\ U(a, t)}{\sum_{a' \in A} \exp \alpha\ U(a', t)} \tag{3}$$

The standard formulation of the RSA speaker policy, as given by Frank and Goodman (2012) and much subsequent work, conforms to this general utility-optimization picture, but is usually stated in terms of a speaker optimizing their linguistic expressions relative to a literal listener, giving rise to the "standard RSA reasoning sequence" of iterated reasoning:

(1) **standard RSA reasoning sequence**:
literal listener → pragmatic speaker → pragmatic listener

Since this literal listener is typically construed as a context-blind, unstrategic dummy, however, it is also possible, if not occasionally more useful (see Section 4), to think of a probabilistic pragmatic model, more generally, as consisting of just two components:

1. a speaker policy that seeks to approximate optimal behavior under some (numeric) utility defining "efficient" or "successful" communication; and

2. a Bayesian listener, who inverts this speaker policy in order to infer likely explanations for the observed linguistic behavior.

Many interesting variations, extensions and applications of this general approach to probabilistic pragmatic modeling have been developed over the years. These often extend the factors that feed into the speaker's utility function, e.g., to include aspects such as politeness or social meaning (e.g. Burnett, 2019; Yoon et al., 2020; Carcassi and Franke, 2023). Another common extension is the inclusion of multiple latent variables for the listener to infer, each of which may shape the speaker's policy in different functional ways. Examples are the inclusion of a question under discussion (Kao et al., 2014) or comparison classes (Graf et al., 2016; Tessler, Tsvilodub, et al., 2020; Tessler and Goodman, 2022). All of these mentioned works and many others tightly link up with experimental data, either for testing categorical hypotheses supported by particular models, or, more frequently, to assess the quantitative fit of one or several models. If done right, this can be one of the assets that probabilistic modeling approaches bring to experimental pragmatics, as argued more extensively in the next Section 3.

Some models and applications also explore, or even require, the higher-order iteration of speaker policies and listener interpretations (e.g. Franke and Jäger, 2014; Bergen, Levy, and Goodman, 2016; Frank, Emilsson, et al., 2016), i.e., an extension of the standard RSA reasoning sequence in (1). The general idea is that a speaker may optimize their linguistic behavior to be informative (or otherwise optimized) for a specific listener interpretation behavior that itself is optimized to the speaker's production policy, and so on. It then becomes an empirical question how much iterated social reasoning is routinely employed by language users in general, or how much iterated perspective-taking individual language users are capable of (e.g., Franke and Degen, 2016). However, from a resource-rational perspective on efficient cognitive processing (Lieder and Griffiths, 2019), we should expect to see meta-cognitive allocation of reasoning effort to be flexibly employed on-demand — a perspective that will be expanded on in Section 4.

3 Theory-driven probabilistic modeling

A now-familiar starting point for any discussion of formal modeling is George Box's dictum that all models are wrong, but some are useful (Box, 1979). Yet acknowledging that no model can be strictly true does not mean that all models are wrong to the same degree. Following Isaac Asimov's observation that wrongness itself admits of degrees—that the belief the Earth is a sphere is less wrong than the belief that it is flat, even though neither is exactly correct—we should prefer models that are less wrong over those that are more so (Asimov, 1988). This raises the natural question of what makes such (less) wrong models useful in the first place.

Just as the familiar principle "garbage in, garbage out" governs the relationship between data quality and the conclusions we can draw from it, an analogous principle governs models themselves: the more (reasonable) assumptions we build in, the more we can conclude from the data. Causal modeling in statistics offers a clear illustration of this logic (Pearl, 2009): by committing to a set of reasonable structural assumptions about how variables relate to one another, we are able to extract substantive, interesting conclusions that would otherwise remain hidden in purely correlational analyses. Central to all of this is the idea that every model encodes, implicitly or explicitly, a set of assumptions about the data-generating process. The richer and more explicit these assumptions are about the underlying process, the more information we are able to extract from the data itself, even though any conclusion must be interpreted as hedged "on the assumption that the model is correct."

Probabilistic pragmatics model have been used, in this way, as statistical models to learn from data, which is likely one of the biggest conceptual contributions of modelling to experimental pragmatics. A prominent class of theory-driven modeling in pragmatics is *latent semantic inference*. Arguably, semantic meanings are not directly observable but must be inferred from indirect evidence. For example, the vague quantifier *many* can convey quite different absolute numbers in sentences like in (3), arguably because its denotation is sensitive to general expectations about the relevant quantities, such as the number of shoes owned, or number of pages read.

(2) Jake owns many pairs of shoes.

(3) Joe read many pages in his book last week.

Fernando and Kamp (1996) hypothesized that, despite this context-dependence, there is still a stable semantic core meaning based on an absolute threshold θ on the cumulative distribution of the expectation about the relevant quantity. So, once we know a person's or a population's expectation of "shoes owned" or "books read last week," we may be able to infer the latent value θ that governs the use of *many* from choice data or acceptability ratings (Schöller and Franke, 2017a; Reese, Jasbi, and Morgan, 2026). In general, latent semantic inference uses data D from an experiment to backwards infer latent parameters θ that influences the semantic meaning of an expression, via a likelihood function $P(D \mid \theta)$ which, in turn, is given by or built around a probabilistic speaker or listener model.

The power that probabilistic pragmatic modeling can unfold if used as part of the data-analytical pipeline is not restricted to parameter inference. Once we acknowledge that a theoretically informed model can become the statistical model itself, multiple models, differing in meaningful theoretical distinctions can then be stringently contrasted through statistical model comparison. For example, van Tiel, Franke, and Sauerland (2021) compared models of quantifier use to distinguish between a standard logical semantics based on thresholds (Barwise and Cooper, 1981) and a semantics in terms of distance to a prototype (McCloskey and Glucksberg, 1978), using a probabilistic pragmatic speaker policy based on informativity-sensitive utilities as a model of the data-generating process with latent parameters inferred for either thresholds or prototypes of quantifiers. Similar methods have been used in the domain of quantifiers (Carcassi and Szymanik, 2021; Reese, Jasbi, and Morgan, 2026; Tiel et al., 2026), vague gradable adjectives or probability expressions Cremers (2022) and van Tiel, Franke, and Sauerland (2022), generics Tessler and Goodman (2019), general (syntactic) ambiguity (Franke and Bergen, 2020), different models of referential reasoning (Qing and Franke, 2015; Franke and Degen, 2016), or intensifiers (Schöller and Franke, 2017b; Zhao, 2019).

4 Towards a computational cognitive science of communication

The blueprint for probabilistic pragmatics sketched in Section 2 suggests that linguistic communication is a pair of mind-*anticipation* (speaker) and mind-*reconstruction*, which is jointly resource-efficient approximately optimal (REAL). This joint-optimization picture has inspired analyses in terms of game-theoretic solutions, e.g., variations on Nash equilibria (e.g., Parikh, 1988) or higher-order iterations of best responses (e.g. Pavan, 2013; Rothschild, 2013; Franke and Jäger, 2014). But while these approaches can tell us which behavioral policies are jointly optimal, they do not give a processing-level account of how human cognition implements (or finds) these strategies.[3] Probabilistic models (in the RSA tradition) are, in principle, better positioned to yield cognitively plausible explanations of the pragmatic reasoning processes involved in context-dependent linguistic communication, as they tend to make leaner assumptions in the sense of REAL behavior and allow to be used as theory-driven statistical models (as argued in Section 3). However, I argue here that in order to develop into an computational cognitive science of communication, probabilistic modeling needs to integrate more cognitive realism (e.g., by borrowing ideas from computational cognitive neuroscience) specifically when it comes to integrating the role of context and the necessity for communicators to reason about the context itself (the very scaffolding in which their behavior takes place). A goal for future expansions of probabilistic modeling in experimental pragmatics should be to formalize how interlocutors reason about others' dynamically-adaptive representations of the communicative situation itself; a form of probabilistic reasoning about context and common ground.[4] In the following, I will spell out this argument by first describing what the prevalent foundational shortcoming is. I then motivate how general aspects of human cognition (with a focus here on attention, awareness, and meta-cognition) are essential to "get REAL" about probabilistic reasoning about context and common ground in linguistic communication.

Both game-theoretic and probabilistic pragmatics in the tradition of the RSA framework are explanatory as models of stylized communication, matching *some*, arguably essential, properties of human linguistic communication at a general level, or matching the specific requirements of carefully designed, artificial laboratory tasks. Nevertheless, they are misspecified as reasonable computational cognitive models of human pragmatic communication in naturally occurring (face-to-face, written ...) communication situations, because they start of from the wrong problem description, i.e., from a wrong model of what context-dependent linguistic communication requires human interlocutors to do, given the general make-up of human information processing. This criticism has an interesting but twisted parallel to early arguments proposed by relevance theory (Sperber and Wilson, 1995; Wilson and Sperber, 2004), namely that linguistic communication is not an (information-theoretic) encoding and decoding problem, but a genuinely *inferential process*. However, in my view, the main difference between (information-theoretic) encoder-decoder architectures and human linguistic communication is not that the latter is inferential, while the former is not (information-theory also standardly uses Bayesian inference as the decoding strategy, just like probabilistic pragmatics does). Rather it is because human communication is based on inference in a massively uncertain, richly structured space, including other minds and their dynamics (as influenced by speech and other actions and events), and uncertainty about what knowledge

[3]This criticism is reminiscent of the motivation behind *relevance theory* (Sperber and Wilson, 1995; Wilson and Sperber, 2004), namely to embed a theory of pragmatic language use in the broader picture of human cognition and its evolution. Yet while I consider what relevance theory wanted to do (namely to see pragmatics in a broader cognitive picture) to be right, I also side with those who see flaws in relevance theory's realization of its ambitions (e.g., Gazdar and Good, 1982; Levinson, 1989).

[4]My present concern is specifically with probabilistic models in experimental pragmatics, but, I believe, that the my more narrow arguments here are kindred to the more general arguments recently developed in the broader context of psycholinguistics by Paula Rubio-Fernandez (**Rubio-Fernandez2024:Cultural-evolutRubio-Fernandez2025:First-acquiring**; Rubio-Fernandez and Harris, 2025, cf.,).

to assume for context-enriched efficient information flow. Standard game-theoretic models like Lewisian signaling games (Lewis, 1969) or probabilistic models in the RSA framework also essentially only provide solutions for an encoding-decoding problem when they assume fixed sets of interpretation options, linguistic actions and receiver responses. This is because they do not factor in the problem that human cognition, outside of stylized laboratory experiments, must operate in an open-ended hypothesis space, which is a problem that language production itself must be catered to mitigate: if efficient language production is "mind-anticipation" then this must entail anticipating "mind-dynamics" in an open-ended hypothesis space of others' mental representations. This implies that future models of pragmatic reasoning should ideally stop being agnostic about the role of core cognitive processes governing "mind dynamics" and instead model REAL linguistic behavior as attuned to general cognitive mechanisms designed to harness open-endedness, most notably attention, awareness, and meta-cognition.[5]

To illustrate the importance of reasoning about attention and awareness, consider an example of written communication as in the following stylized exchange:

(4) Sender: If I were a ~~German~~ person who didn't like spicy food, I'd not go there.
Receiver (a German): Touché!

This is an example of a general and quite productive class of pragmatic inferences, which I here call *strike-through implicature*.[6] The writer chose to explicitly mark an expression as deleted or to be ignored, in full awareness that the reader will recognize that the speaker has thereby made the reader aware of the speaker's as-if initial consideration of the elided text. This leads to a picture of asymmetric availability of alternative utterances: an utterance of (4) with strike-through may signal that the speaker has considered both of the alternatives (5) below, ended up preferring (5b) but also wants the recipient to know that they have deliberately NOT chosen the alternative (5a).

(5) a. If I were a German person who didn't like spicy food, I'd not go there.
b. If I were a person who didn't like spicy food, I'd not go there.

Most importantly here, *if* the speaker had only written (5b), the recipient would never even have considered that the speaker deliberately did NOT chose (5a). Much more could be said about strike-through implicatures and like-minded cases of pretend-errors in spoken language (think: "Whoops!? Did I really just say that?"), but the key take-away for the present argument is, quite general and imprecisely stated, that pragmatic inferences draw on the complex, counterfactual dynamics of awareness and attention that speakers can anticipate and listeners can recover. Some work in formal semantics and pragmatics has explored effects of awareness or attention dynamics (e.g., de Jager, 2009; Kaufmann, 2009; Crone, 2019; Klecha, 2018; Westera, 2022) and while there are logical (e.g., Fagin and Halpern, 1988; Halpern, 2001; Heifetz, Meier, and Schipper, 2006, 2008) and game-theoretic (e.g., Feinberg, 2004; Heifetz, Meier, and Schipper, 2007; Franke, 2014)

[5]Pragmatic reasoning must also be optimized to general properties of human memory, so that, e.g., a speaker can include reminders and properly manage what's individually known, shared information, and what's new and important. Common speech-practices are clearly attuned to these (temporal) properties of memory, attention and awareness: since people forget, we have practices of reminding and even joint reminiscing; since people are inattentive, we know that simple reminders work and are commonly recognized, as reflected, e.g., by the fact that "Thanks." can be the right reaction to a question like "Did you call Jake?"

[6]Strike-through implicatures in written communication are likely a cousin to or a special case of apohasis. As far as I can tell, neither strike-through implicatures nor other kinds of "pretend-errors" have been treated systematically in the pragmatic literature (possibly because they would require reasoning about attentional dynamics).

frameworks for modeling reasoning about unawareness, inclusion of attention dynamics into a REAL cognitive processing model for pragmatic communication is missing.

The next example tries to motivate the importance of considering meta-cognitive processes that may steer awareness dynamics in pragmatic reasoning (see Allott, 2020, for further explorations of the importance of meta-cognition on pragmatic reasoning). Cummins and Franke (2021) discuss the argumentative use of quantifying expressions based on selected cases of UK universities' self-reporting on the results of the 2014 Research Excellence Framework (REF), an example of which is (6).

(6) Royal Holloway is within the top 25 per cent of United Kingdom universities for research rated 'world-leading' or 'internationally excellent'.

Pragmatic interpretation of this statement can vary in terms of awareness and depth of reasoning about the strategic context. For illustration, processing of (6) could proceed along a path of increasing effort of contextual reasoning like this:

(7) a. take the statement to be a simple informative utterance about some aspect of Royal Holloway's REF results;
  b. realize that this statement is likely intended as some kind of argument for *something* (without knowing what);
  c. take the statement to be an argument for some specific conclusion, such as the idea that Royal Holloway is a good university, according to the REF;
  d. realize that the argument made is rather weak, thus suggesting the opposite conclusion.

Notice that interpretative depth, as suggested by (7), does not amount to iterated social reasoning, but the degree to which the wider strategic context of utterance production is taken into account. Intuitively, reaching a certain reasoning level, e.g., up to (7c), may be effortless and automatic, while passing on to a deeper level of contextual reasoning, such as up to (7d), may require more controlled allocation of attentional resources. If that is so, then part of REAL pragmatic reasoning consists in a (meta-cognitive) decision of whether a simpler reasoning level is good enough. As with other uncertain variables, to decide whether to update or change a context representation, agents would therefore need to track meta-cognitive confidence (Fleming, 2024) about the level of depth or granularity with which they represent the relevant aspects of the context. Uncertainty about context representations and the tracking of meta-cognitive confidence to exert cognitive control can, in principle, be included in probabilistic pragmatics models of the RSA variety, e.g., by including two-stage Bayesian models of confidence (Fleming and Daw, 2017; Schulz, Fleming, and Dayan, 2023). The most important point for present concerns is, however, that *if* aspects of meta-cognition matter for the pragmatic interpreter, then they also likely matter for language production, since — in strategic contexts with a certain misalignment of interests— speakers might then optimize their behavior (implicitly) towards not raising suspicion, i.e., anticipating how (Bayesian) confidence is calculated and optimizing production to minimize the listener's chance of becoming aware of particular features of the context. So, including meta-cognitive revision of beliefs about the context-model likely also contributes to a better understanding of non-canonical (e.g., not fully cooperative) cases of language use.

Further directions for a dedicated cognitive turn in modeling pragmatic reasoning include factors like learning and adaptation over the course of an experiment (e.g., Schuster and Degen, 2020) or with hierarchical generalization to a larger population (Hawkins, Franke, et al., 2023). This is especially interesting also when integrated into a general framework for modeling learning and decision-making, like the ACT-R cognitive architecture (J. R. Anderson, 1993; J. R. Anderson et al., 2004), as in recent work by Duff, Mayn, and Demberg (2026). A more thorough cognitive orientation of model-centric pragmatics also seems called for to deal with REAL behavior in temporally extended communicative situations, such as the incremental choice of an utterance or multi-turn conversations. For instance, recent models for the incremental production of referential expressions like "the big blue box" treat each modifier as a potentially individual choice point for the speaker, to be selected sequentially based on utilities that are dynamically updated after the incremental interpretation effects of a partial utterance have been taken into account (Cohn-Gordon, Goodman, and Potts, 2019; Schlotterbeck and Wang, 2023). The plea for more cognitive realism in probabilistic modeling, when applied to (especially: face-to-face) referential communication, leads directly to a view also advocated by Rubio-Fernandez, Berke, and Jara-Ettinger (2025), namely to explicitly attend to what the authors call "social micro-processes," which are (very) REAL solutions for coping with the complexities of iterated social reasoning in the sense introduced and argued for here. As for more extended dialogue situations, some recent work has looked at the selection of questions and answers (Hawkins, Stuhlmüller, et al., 2015; Hawkins, Tsvilodub, et al., 2025), or at multi-turn conversations to coordinate on opinions (Achimova, Scontras, et al., 2022; Achimova, Franke, and Butz, 2025), and even at extended dialogues (Das et al., 2026). Yet, while illuminating pioneering work exists (Woensdregt, Cummins, and Smith, 2020; C. J. Anderson, 2021; Qing, Goodman, and Lassiter, 2016), general, cognitively plausible models of REAL pragmatic reasoning in longer conversations, especially when it comes to probabilistic reasoning about common ground, remain an exciting future challenge.

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